Bryan Wei

LinkedIn GitHub Portfolio l76wei@uwaterloo.ca

Skills

Languages: Python, TypeScript, JavaScript, Java, C#, C++, C, SQL, Bash, PowerShell

Frameworks: React, Next.js, React Native, three.js, Node.js, Express, FastAPI, Spring Boot, PyTorch

Tools & Databases: Git, Linux, AWS, Azure, Unity, Bun, PostgreSQL, MySQL, SQLite, MongoDB, Convex, Tinybird

Education

University of Waterloo, *Honours Computer Science Co-op* *Sep 2024 – Present*

Experience

Full Stack Developer *Sep 2026 – Present*
Workplace Safety and Insurance Board – Innovation Lab *Waterloo, ON*

- Engineered a **multithreaded synthetic test-data engine** that writes schema-configurable datasets to **SQL** files and **Microsoft Fabric** tables, cutting **1M-row** generation time by **65% (20s → 7s)**
- Built an **indoor wayfinding app** helping **3,500+ employees** across **8 offices** learn their workspaces during RTO, using **QR-anchored positioning** and **Dijkstra routing** over multi-floor maps

Software Developer *Sep 2026 – Present*
Waterloo Reality Labs *Waterloo, ON*

- Developed a **few-shot Siamese network** using **PyTorch triplet loss** that recognizes new hand gestures from only **1–3 reference clips** at up to **98% accuracy**, deployed via **ONNX** for in-engine inference in **Unity**
- Designed a **gesture intensity scoring module** that rates hand-gesture effort on a normalized 0–1 scale from **kinematic features** (joint velocity, snap, movement time), achieving ≥**90% ranking accuracy** across effort levels

Software Developer *May 2026 – Aug 2026*
Traces *Toronto, ON*

- Built core features for Traces, a developer platform used by **100+ organizations** to capture, share, and continue **AI coding-agent** sessions across tools and teams
- Implemented **data ingestion adapters** for **14+ coding agents**, handling local session discovery, content parsing, and **schema normalization** across **JSONL** files, **SQLite** databases, and cloud-backed session stores
- Engineered a **subagent discovery pipeline** that infers parent–child relationships for **8+ agents** with **97% accuracy**, preserving lineage across local indexing, API surfaces, remote sync, and **Convex** storage
- Developed **token usage tracking** for **8+ agents** at **99% accuracy**, mapping agent-specific per-API-call and session-level usage into shared **TypeScript** schemas, with CLI sync and persistence in **Convex** and **Tinybird**
- Designed a provider-neutral **cloud agent session capture engine** spanning polling, webhook, and streaming interfaces, with **concurrent job processing** and **idempotent, retry-safe persistence**

Projects

StickDrift *Oct 2026*

- Developed a **movement disorder monitoring platform** for seniors using **React**, **TypeScript**, **three.js**, **FastAPI**, and **PostgreSQL**, where senior-oriented controller games capture joystick and accelerometer data to detect early signs of movement disorders with **machine learning**
- Trained a **PyTorch Temporal Convolutional Network** on joystick and accelerometer **time series** to detect tremor, stroke, and Parkinson's movement patterns with **96% accuracy**, feeding pattern scores into a caregiver dashboard
- Built a **real-time telemetry pipeline** streaming controller input over **FastAPI WebSockets**, enabling caregivers to monitor movement live

Contour *Sep 2026*

- Built a **project visualization platform** using **React**, **TypeScript**, **Node.js**, **Express**, and **PostgreSQL**, automatically transforming GitHub activity and coding-agent sessions into an interactive project map
- Engineered **GitHub webhook** ingestion for PRs, reviews, and CI runs, plus a local collector that scans **Codex**, **Claude Code**, and **Cursor** sessions and **incrementally syncs** new activity via **SQLite** checkpoints
- Designed an **asynchronous event-processing pipeline** on **pg-boss** job queues and **PostgreSQL**, using the **OpenAI API** to organize raw development activity into the live project map